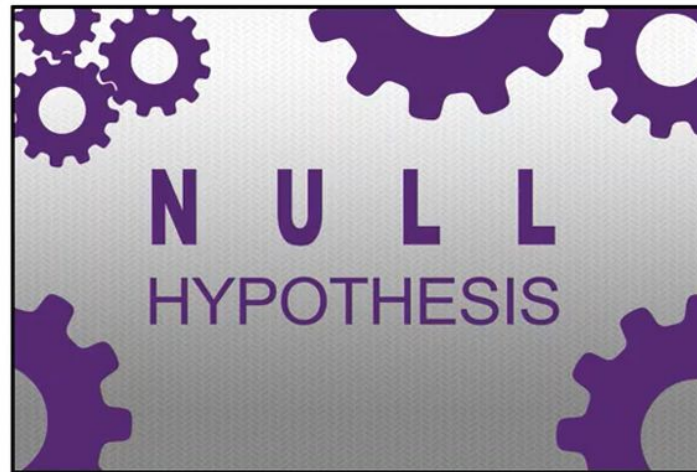


w2.1- Foundational Principles in Hypothesis Testing (6:41)

w2.1- Foundational Principles in Hypothesis Testing (6:41).mp4

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Null Hypothesis



00:00:37 / 00:06:41

55

Test Statistic



There are two ways to test the hypothesis.

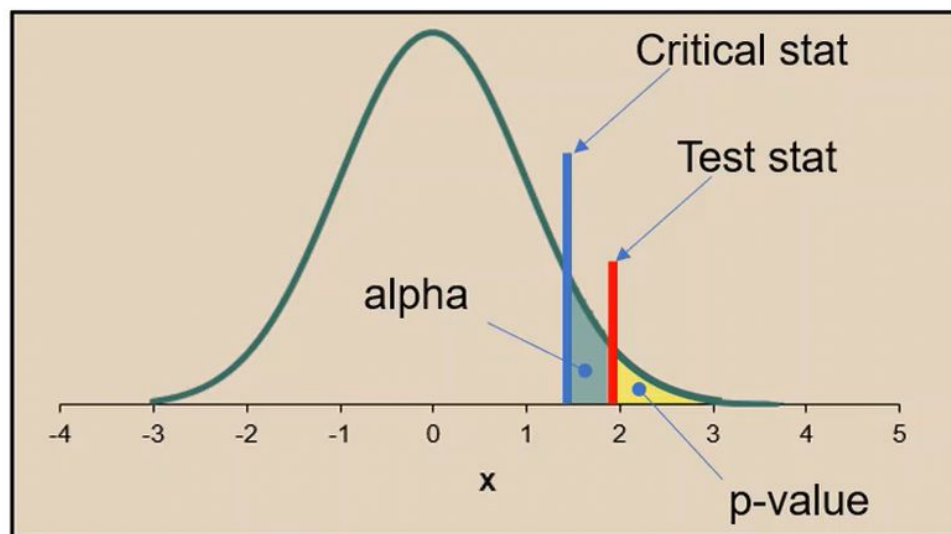
Types of Errors

	Null is True	Null is False
Decide to Fail to Reject the Null	OK	Type II Error Beta- Consumer False -ve
Decide to Reject the Null	Type I Error Alpha False +ve	OK

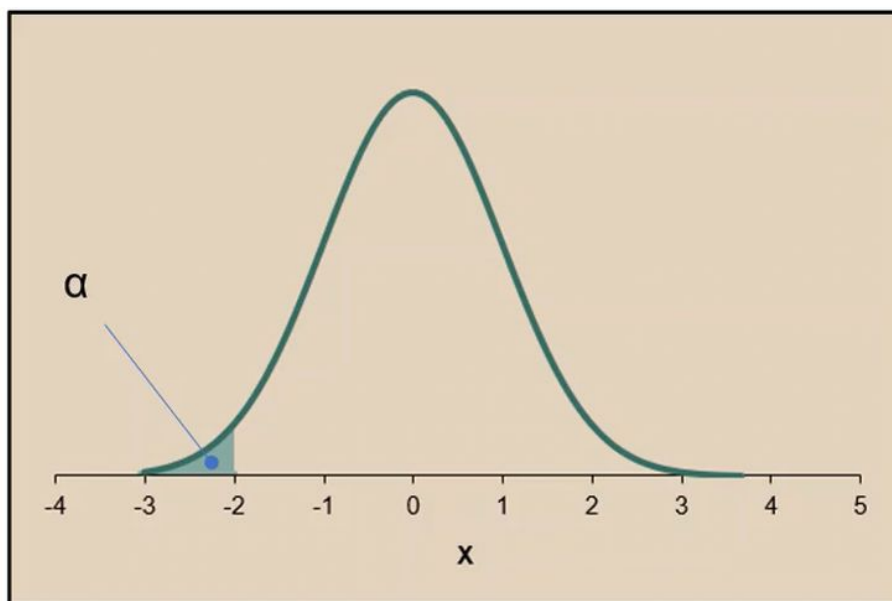
risk



P-Value



One Tail Test



$$H_0: \mu \geq 5$$

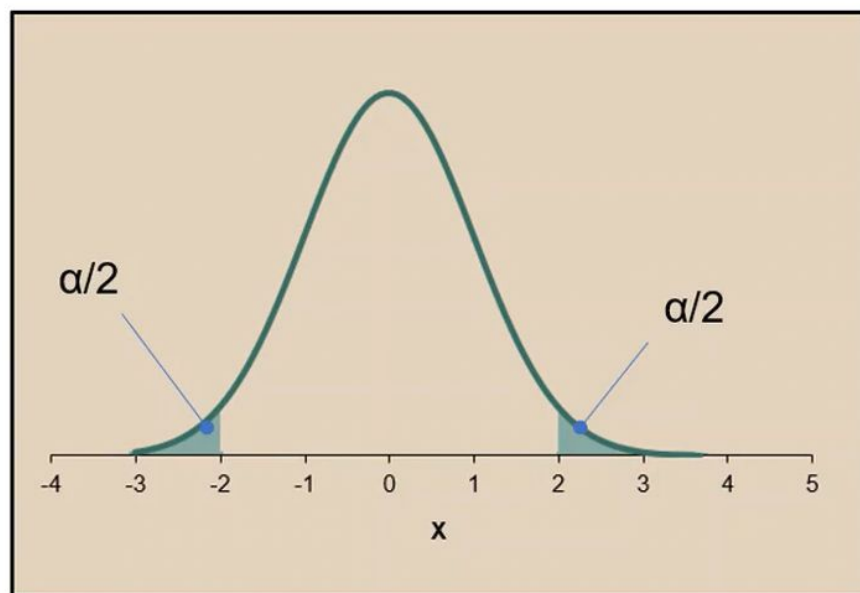
$$H_1: \mu < 5$$

If a null hypothesis is established to test whether a sample value is smaller or



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Two Tail Test



$$H_0: \mu = 5$$

$$H_1: \mu \neq 5$$

If a null hypothesis is established to test whether a sample value is different



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1. Two-Tailed Test (Non-Directional)

A two-tailed test is used when you want to determine if there is **any significant difference** between groups, but you don't necessarily care (or know) which one is larger or smaller.

- **Hypothesis:** $H_a : \mu \neq \text{target}$
- **The "Split":** The significance level (α) is split between both ends of the distribution. If $\alpha = 0.05$, then 2.5% of the risk is in the left tail and 2.5% is in the right tail. 
- **Standard:** This is the "default" in most scientific research because it is more conservative. You are testing for both "much greater than" and "much less than."

2. One-Tailed Test (Directional)

A one-tailed test is used when you are only interested in a change in **one specific direction**.

You are testing if a value is strictly "greater than" or strictly "less than" a target. 

- **Hypothesis:** $H_a : \mu > \text{target}$ OR $H_a : \mu < \text{target}$
- **The "Concentration":** The entire significance level (α) is placed in one tail. If $\alpha = 0.05$, the entire 5% is at one end.  
- **Power:** This test has more statistical power to detect an effect in the chosen direction because the critical value is easier to reach. However, if the result goes in the *opposite* direction, you must fail to reject the null, even if the difference is huge. 

This transcript provides a comprehensive overview of **Hypothesis Testing** within the Six Sigma Black Belt framework. Below is a structured summary of the key terminology and concepts to help you master these statistical foundations.

1. The Core Hypotheses

In any statistical test, we weigh two mutually exclusive statements:

- **Null Hypothesis (H_0):** The "status quo" or the statement assumed to be true until proven otherwise. It **always includes equality** ($=$, $\leq$, or $\geq$).
 - *Analogy:* In a legal trial, H_0 is "not guilty."
- **Alternative Hypothesis (H_a or H_1):** what we hope to prove or find evidence for. It represents a difference, change, or effect.

Key Takeaway: We never prove the null hypothesis is "innocent"; we only **reject** it or **fail to reject** it based on the evidence.

2. Decision Criteria: How We Test

There are two primary methods for deciding whether to reject the null hypothesis:

Method A: Test Statistics vs. Critical Statistics

- **Test Statistic:** Calculated from your sample data.
- **Critical Statistic:** A "threshold" value determined by your chosen risk level (α).
- **The Rule:** If $|\text{Test Statistic}| > |\text{Critical Statistic}|$, **Reject H_0** .

Method B: P-value vs. Alpha (α)

- **Alpha (α):** The threshold for significance (usually 0.05). It is the area in the tail(s) of the distribution.
- **P-value:** The actual probability of seeing your sample data if the null hypothesis is true.
- **The Rule:** If **P-value** $< \alpha$, the result is significant $\rightarrow$ **Reject H_0** .

3. Error Types: When Inferences Go Wrong

Because we use samples to describe populations, there is always a risk of error:

Error Type	Common Name	Definition	Probability
Type I	Producer's Risk	Rejecting H_0 when it is actually true (False Alarm).	α (Alpha)
Type II	Consumer's Risk	Failing to reject H_0 when it is actually false (Missed Opportunity).	β (Beta)

Note on Trade-offs: For a fixed sample size, α and β are inversely related. To reduce both risks simultaneously, you must **increase the sample size.**

4. One-Tailed vs. Two-Tailed Tests

The structure of your alternative hypothesis determines where the "rejection region" lies on the distribution curve.

One-Tailed Test

Used when you are looking for a specific direction (e.g., "Is the new process *faster*?").

- **Rejection Region:** Entirely in one tail (left or right).
- **Example:** $H_a : \mu > \text{Value}$

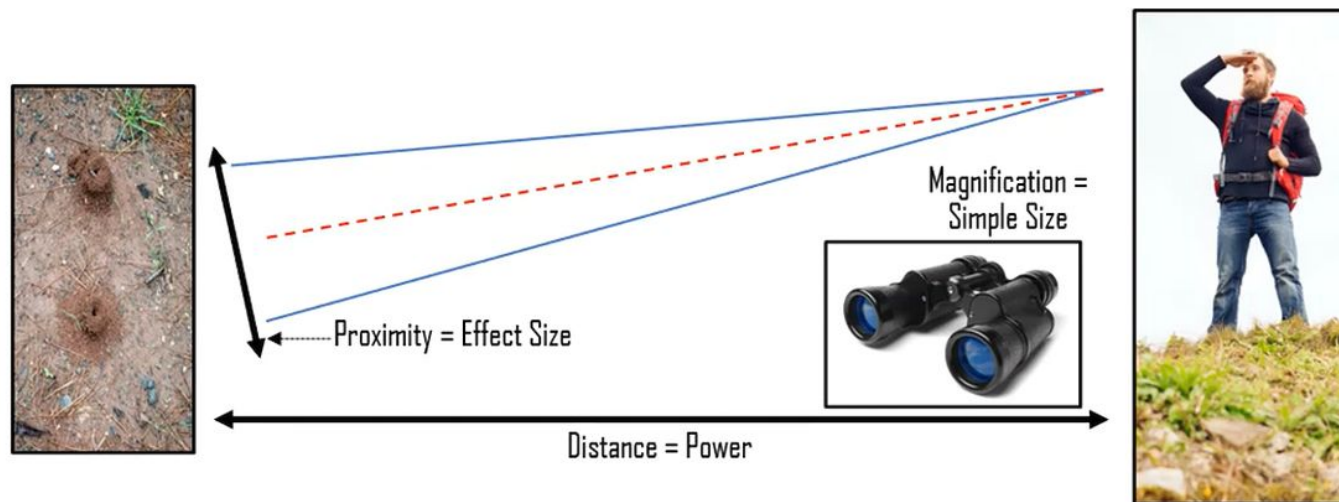
Two-Tailed Test

Used when you are looking for *any* difference (e.g., "Is the *weight different* than the target?").

- **Rejection Region:** Split equally between both tails.
- **Note:** Because α is split ($\alpha/2$ in each tail), the critical statistics move further away from the center, making the test more *conservative.*

w2.2- Effect Size and Power (8:50)

Power Terminology



Let's define three parameters,

Effect Size Settings

Parameter	Small	Medium	Large
ρ (correlation)	0.1	0.3	0.5
Cohen's d	0.2	0.5	0.8
d_z	0.2	0.5	0.8
ρ^2 (regression)	0.02	0.13	0.26
Cohen's h (proportion)	0.2	0.5	0.8
f	0.1	0.25	0.4
η^2	0.0099	0.0588	.1379
Hedges g	0.2	0.5	0.8
ω^2	0.0099	0.0588	.1379

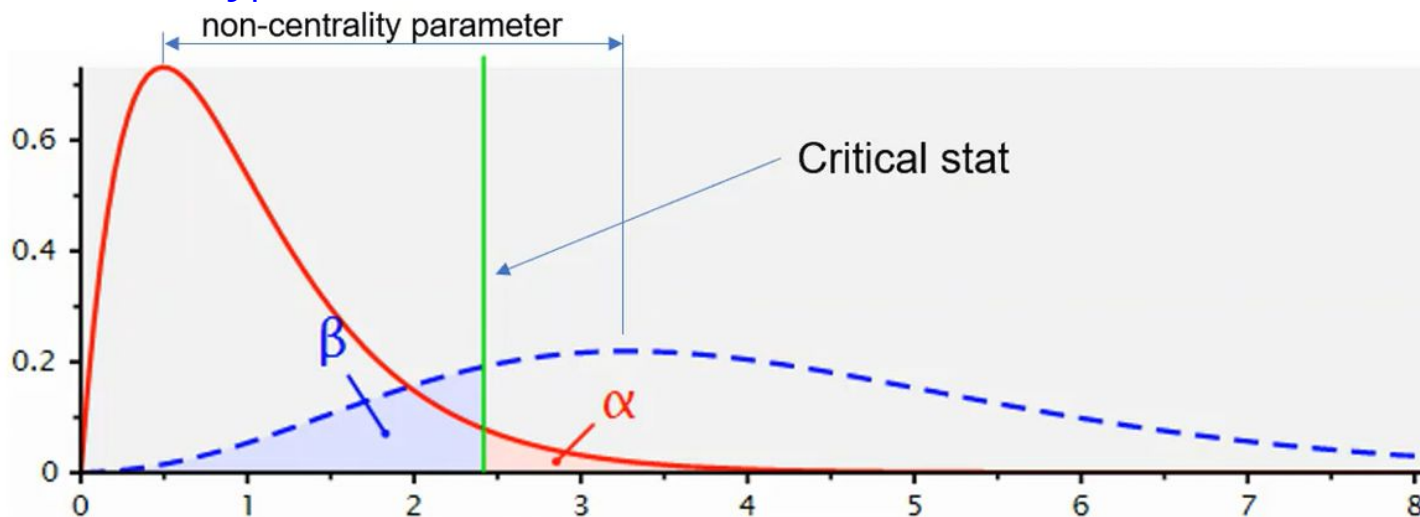


w2.2- Effect Size and Power (8:50).mp4

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G-Power

Null hypothesis



<http://www.gpower.hhu.de/>

statistical power analysis
for many different t-test,



Non-Parametric Power

Parametric Test	Non-Parametric Test	Efficiency
T Test (Paired Data)	Sign Test	63.66% ($2/\pi$)
T Test (Paired Data)	Wilcoxon Signed Ranked Test	95.49% ($3/\pi$)
T Test (Independent Data)	Wilcoxon rank sum test	95.49% ($3/\pi$)
ANOVA	K-W Test	95.49% ($3/\pi$)
R-ANOVA	Friedman Test ($k=3$)	72% ($3k/\pi(k+1)$)
Pearson Correlation	Spearman	91.19% ($9/\pi^2$)

power analysis is the role
of the parametric test.



Non-Parametric Power Example

Pearson Correlation: Power = $1 - \beta = 0.8$, Minimum Sample = 84

Spearman Correlation: Actual Power = $0.8 * 0.9119 = 0.7295$

Equivalent Power = $0.8 / 0.9119 = 0.8773$

Adjusted Minimum Sample = 104

As an example, if a
Pearson correlation with



This section dives into the critical relationship between **Statistical Power**, **Effect Size**, and **Sample Size**, while distinguishing between statistical and practical significance.

1. Significance: Statistical vs. Practical

Not all significant results are useful.

- **Statistical Significance:** Tells you if an effect exists beyond random chance (determined by α and p-values).
- **Practical Significance:** Asks if the effect is large enough to care about in the real world.
 - *Example:* A process change might statistically reduce defects by 0.001%, but if the implementation costs \$1M, it lacks **practical significance.**
- **The Overpowering Risk:** Using an excessively large sample size can make even tiny, meaningless differences appear "statistically significant."

2. The Anthill Analogy

To understand the mechanics of power analysis, visualize standing in a field looking at what might be one or two anthills:

- **Power (Distance):** How far away you are standing. Usually set at **0.8**. High power means you are less likely to miss an effect that is actually there.
- **Effect Size (Proximity of Anthills):** How far apart the two anthills are. If they are right next to each other (small effect), they are harder to tell apart.
- **Sample Size (Binocular Magnification):** The tool you use to see more clearly. If the effect size is small, you need "stronger binoculars" (a larger sample size) to maintain your power.

3. Measuring the "Effect"

Different tests require different measures to quantify the magnitude of an effect:

Measure	Application	Context
Cohen's d	Differences between two means	Standardized difference.
Cohen's f	ANOVA / F-distribution	Used for more than two groups.
Eta Square (η^2)	ANOVA	Proportion of variance explained; can be biased.
Omega Square (ω^2)	ANOVA	A more robust, less biased version of Eta Square.
Hedges' g	Small samples	Better than Cohen's d when $n < 20$.
Rho (ρ)	Correlation	Measures strength of relationship.

4. G-Power and Distribution Curves

G-Power is the industry-standard tool for calculating necessary sample sizes. It works by looking at the overlap of two distributions:

1. **The Central Distribution (Red)**: Represents the Null Hypothesis (H_0).
2. **The Non-central Distribution (Blue)**: Represents the Alternative Hypothesis (H_a).

The goal: Minimize the area where they overlap **too much (Beta)**, thereby maximizing **Power** ($1 - \beta$). The distance between the peaks of these two curves is the **non-centrality parameter**, which helps determine your required sample size.

5. Parametric vs. Nonparametric Power

- **Parametric tests** (like Pearson correlation) are more "powerful"—they require fewer samples to reach a 0.8 power level.
- **Nonparametric tests** (like Spearman correlation) are safer when data isn't normal, but they have less "punch."
- **The Penalty:** If you switch to a nonparametric test, you must increase your sample size to maintain the same level of statistical power. As shown in your transcript, a study requiring 84 samples for a Pearson test would need 104 samples for a Spearman test to achieve the same confidence.

The choice between parametric and non-parametric tests usually comes down to how much you know—or can assume—about the "shape" of your data.

1. Parametric Tests

Parametric tests assume that your data follows a specific distribution, usually the **Normal (Gaussian) Distribution** (the "bell curve"). They rely on parameters like the **mean** and **standard deviation**.  

- **Requirements:** Data should be continuous (interval or ratio scale), normally distributed, and have similar variances across groups. 
- **Strengths:** They have high **statistical power**, meaning they are more likely to detect a real effect if one exists. 
- **Examples:** t-tests, ANOVA, Pearson Correlation.

2. Non-Parametric Tests

Non-parametric tests are often called "**distribution-free**" tests because they do not assume your data follows a normal curve. Instead of using the raw values (like the mean), they typically use the **ranks** of the data.  

- **Requirements:** Very few. They work well for ordinal (ranked) data, small sample sizes, or data with significant outliers. 
- **Strengths:** They are **robust**. If your data is skewed or "messy," these tests won't give you misleading results like a parametric test might.
- **Examples:** Mann-Whitney U, Wilcoxon Signed-Rank, Kruskal-Wallis, Spearman's Rho.

w2.3- Estimators and Confidence Intervals (5:26)

Sample Size

$$n = \frac{Z^2 \sigma^2}{E^2}$$

Where

Z = critical statistic at a given confidence

σ = standard deviation

$\sigma^2 = \hat{p}(1 - \hat{p})$ (for proportions)

E = margin of error

99% confidence $\Rightarrow \alpha=0.01 \Rightarrow Z = 2.575 \Rightarrow 883$ samples



Estimators

- Point estimators
- Interval estimators

$$\text{point estimate} \pm \text{critical stat} * \frac{\text{standard deviation}}{\sqrt{\text{sample size}}}$$

- Unbiased estimators
 - Mean
 - Proportion
 - Variance

When attempting to characterize a data set, we have two types of estimators.



Confidence Intervals for the Mean

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \text{ (large sample sizes)}$$

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \text{ (small sample sizes)}$$

selecting the correct formula comes
down to a handful of criteria.



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Confidence Intervals for Variation

$$\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}$$

Where

n = the sample size

s^2 = point estimate of variance

$\chi_{\frac{\alpha}{2}}^2$ and $\chi_{1-\frac{\alpha}{2}}^2$ = chi-square statistics



This is not true for variance.

Confidence Intervals for Proportion

$$\hat{p} - Z_{\frac{\sigma}{2}} \sqrt{\frac{p(1-p)}{n}} < p < \hat{p} + Z_{\frac{\sigma}{2}} \sqrt{\frac{p(1-p)}{n}}$$

Where

$\hat{p}$ = proportion estimate

n = the sample size

$Z_{\frac{\sigma}{2}}$ = the normal statistic



This final module bridges the gap between hypothesis testing and **Interval Estimation** focusing on how to calculate the sample sizes required for confidence and how to build intervals for different population parameters.

1. Calculating Minimum Sample Size (n)

Before starting a study, you must determine how many samples are needed to meet your desired confidence level and precision (Margin of Error).

For a Population Mean:

The formula derived from the confidence interval is:

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

- $Z_{\alpha/2}$: The critical value (e.g., 1.96 for 95% confidence).
- σ : The standard deviation.
- E : The Margin of Error (the "plus or minus" amount).

For a Population Proportion:

If you are dealing with percentages (proportions), σ^2 is replaced by $p(1 - p)$:

$$n = \frac{Z_{\alpha/2}^2 \cdot p(1 - p)}{E^2}$$

Critical Note: Sample size calculations for confidence intervals are **always two-tailed**. If you want 99% confidence, your α is 0.01, which must be split into 0.005 for each tail to find your Z score.

2. Point vs. Interval Estimators

- **Point Estimator:** A single value (like $\bar{x}$ or $\hat{p}$). While precise, it is almost certainly "wrong" in that it rarely matches the true population parameter exactly.
- **Interval Estimator:** Creates a range (Confidence Interval) around the point estimate. It accounts for sampling error and gives a probability that the true parameter lies within that range.

3. Confidence Intervals (CI) for Unbiased Estimators

We focus on three primary population parameters: **Mean**, **Variance**, and **Proportion**.

A. Confidence Interval for the Mean (μ)

Your choice of formula depends on whether you know the population standard deviation (σ):

1. **Known σ (Normal Distribution):** Use the Z -statistic.

$$\bar{x} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

2. **Unknown σ (Student's t-Distribution):** Use the t -statistic and sample standard deviation (s). This is the safer, **more conservative choice** especially for small samples ($n < 30$).

$$\bar{x} \pm t_{\alpha/2, df} \left(\frac{s}{\sqrt{n}} \right)$$

- *Degrees of Freedom (df):* $n - 1$

B. Confidence Interval for Variance (σ^2)

Unlike the mean, **variance is not symmetrical**. It uses the **Chi-Square (χ^2) distribution**, which is skewed.

The interval is calculated as:

$$\frac{(n-1)s^2}{\chi_{\alpha/2, df}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{1-\alpha/2, df}^2}$$

C. Confidence Interval for Proportion (p)

When your data is attribute-based (pass/fail), you use the normal approximation, provided the sample is large enough ($np \geq 5$ and $n(1 - p) \geq 5$).

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Summary Checklist for Selection

If you are estimating...	And you know...	Use this Distribution
Mean	σ (Population SD)	Z (Normal)
Mean	s (Sample SD only)	t (Student's)
Variance	n and s^2	χ^2 (Chi-Square)
Proportion	n and $\hat{p}$	Z (Normal)